

Exam 3 Info

Exam 3 will be posted on Piazza at 8am on Friday April 28 at 8am and due on Gradescope at 10am on Saturday April 29. The exam will be open book/notes/web resources but NO consultation with any living person whatsoever is allowed in any way, including posting any course-related questions on any websites. (You will, however, be able to post clarifying questions to Instructors on Piazza.)

You should be familiar with all definitions and terminology used in class.

Studying suggestion: Look at all examples covered in lectures, worksheets and homework. For the textbook, see notes on the different sections below:

Section 35 - Div/Mod (*don't read, all we needed was the definition of mod as a binary operation rather than an equivalence relation.*)

Section 36 - gcd, Euclid's Algorithm

Section 37 - Modular Arithmetic (*uses a different notation. Most relevant material is Thm 37.14 and onwards, about inverses.*)

Section 43 - Fermat's Little Theorem / Euler's Theorem (*Don't read this section, the proofs are different from the ones we used in class*)

Section 44 - Intro to Cryptography

Section 46 - RSA

Section 47 - Intro to Graphs

Section 48 - Subgraphs

Section 49 - Connection

Section 50 - Trees

The exam will as usual contain Computations/Short Answers, and Justifications/Proofs:

COMPUTATIONS/SHORT ANSWERS

1. Euclid's Algorithm

- (a) Use Euclid's algorithm to compute $\gcd(a, b) = d$. Ex: Use Euclid's algorithm to compute $d = \gcd(78, 30)$. You don't have to show all steps of the computer program; anything that shows the sequence of gcds you would compute is sufficient.

Solution:

(Writing just the equations is sufficient, and also useful for 1b)

$$78 = 30 \cdot 2 + 18$$

$$30 = 18 \cdot 1 + 12$$

$$18 = 12 \cdot 1 + 6$$

$$12 = 6 \cdot 2 + 0$$

$$\gcd(78, 30) = 6$$

- (b) Use Euclid's algorithm to solve $ax + by = d$. Ex: Use your results of the previous problem to find x and y such that $d = 78x + 30y$.

Solution:

Solving for remainder:

$$18 = 78 - 30 \cdot 2$$

$$12 = 30 - 18 \cdot 1$$

$$6 = 18 - 12 \cdot 1$$

Plugging in:

$$6 = 18 - 12 \cdot 1$$

$$= 18 - (30 - 18 \cdot 1) \cdot 1$$

$$= -30 + 18 \cdot 2$$

$$= -30 + (78 - 30 \cdot 2) \cdot 2$$

$$6 = 78 \cdot 2 + 30 \cdot -5$$

2. Modular arithmetic

(a) *Addition, subtraction, multiplication and powers mod n.* Ex: Fill out charts for addition, multiplication and powers mod 8. (Note: I would only ask for a couple of rows).

(b) *Find inverses mod n (from using Euclid's algorithm to solve $ax + by = d$) if they exist.* Ex: Can you use the results of a previous problem to find the inverse of 30 (mod 78)? Also see RSA example below.

Solution:

No, the equation in 1b) only tells us $6 = 30 \cdot -5 \pmod{78}$, but we need to solve $30b = 1 \pmod{78}$. In fact 30 does not have an inverse mod 78 because it is not relatively prime to 78.

(c) *Find inverses mod n (if they exist) from a mod n table, or by guess-and-check.* Ex: What is the inverse of 5 (mod 8)? What is the inverse of 3 (mod 10)?

Solution:

The inverse of 5 mod 8 is 5.

The inverse of 3 mod 10 is 7.

3. Fermat's Little Theorem

Apply the formula $a^{p-1} = 1 \pmod{p}$ (where p is prime and $a \neq 0$). Ex:

(a) Find a number x such that $4^x = 1 \pmod{23}$, or explain why it is not possible. **Solution:**

$$4^{22} = 1 \pmod{23}$$

(b) Find a prime p such that $7^{20} = 1 \pmod{p}$, or explain why it is not possible. **Solution:**

Note p cannot be 21 since 21 is not prime. But as long as $7^4, 7^5$ or $7^{10} = 1$ we will also have $7^{20} = 1$, so for example when $p = 11$ Fermat's tells us $7^{10} = 1$ and thus $7^{20} = 7^{10} * 7^{10} = 1$.

4. Euler's Theorem

(a) Compute $\mathbb{Z}_n^*$. Ex: Compute $\mathbb{Z}_8^*$. **Solution:**

$$\{1, 3, 5, 7\}$$

(b) Compute $\varphi(n)$. Ex: Compute $\varphi(8)$. **Solution:**

$$4$$

- (c) Apply the formula $a^{\varphi(n)} = 1 \pmod n$ (where $n > 0$ and a is relatively prime to n .) Exs: Can you find a number x such that $3^x = 1 \pmod 8$? Can you find a number x such that $6^x = 1 \pmod 8$?

Solution:

We can find a number x such that $3^x = 1 \pmod 8$:

$$3^4 = 1 \pmod 8$$

We can't find a number x such that $6^x = 1 \pmod 8$ since 6 isn't relatively prime to 8.

5. RSA

- (a) Compute a complete RSA example. Ex:

- $p = 17$

- $q = 13$

- $n = \underline{\hspace{2cm}}$ **Solution:**

$$n = 17 \cdot 13 = \boxed{221}$$

- $\varphi(n) = \underline{\hspace{2cm}}$ **Solution:**

$$\varphi(n) = 16 \cdot 12 = \boxed{192}$$

- $e = \underline{\hspace{2cm}}$ Choose one of these two options for e : 15, 25. Only one will work. Why doesn't the other one work? **Solution:**

$e = 25$. 15 is not relatively prime to 192.

- $d = \underline{\hspace{2cm}}$ Compute d by hand using Euclid's algorithm. (Longest step). **Solution:**

$$d = 169.$$

Computing d

Euclid's algorithm equations:

$$192 = 25 \cdot 7 + 17$$

$$25 = 17 \cdot 1 + 8$$

$$17 = 8 \cdot 2 + 1$$

$$8 = 1 \cdot 8 + 0$$

Solving for remainder:

$$17 = 192 - 25 \cdot 7$$

$$8 = 25 - 17 \cdot 1$$

$$1 = 17 - 8 \cdot 2$$

Plugging in backwards:

$$1 = 17 - 8 \cdot 2$$

$$= 17 - (25 - 17 \cdot 1) \cdot 2$$

$$= 25 \cdot -2 + 17 \cdot 3$$

$$= 25 \cdot -2 + (192 - 25 \cdot 7) \cdot 3$$

$$1 = 192 \cdot 3 + 25 \cdot -23$$

Thus $d = -23$, and choosing the smallest positive representative (easiest for taking powers) gives us $d = 169 \pmod{192}$.

- Bob is constructing this example so Alice can send him messages. What is the public key he will send to Alice? **Solution:**

(221, 25)

- What is Bob's private key? **Solution:**

(221, 169)

- Alice wants to send the message $M = 201$. What does she compute to send the encrypted message N ?
 $N = E(M) = \underline{\hspace{2cm}}$. (should be a numerical expression, or compute using WolframAlpha if allowed on exam.) **Solution:**
 $N = E(M) = 201^{25} \bmod 221$
- What does Bob do to get M back?
 $M = D(N) = \underline{\hspace{2cm}}$. (should be an expression in terms of N , or compute using WolframAlpha if allowed on exam.) **Solution:**
 $M = D(N) = N^{169} \bmod 221$

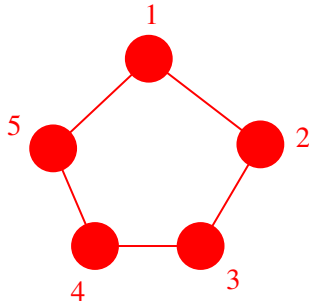
6. Graphs

Note: K_n = complete graph on n vertices, C_n = cycle on n vertices, P_n = path graph on n vertices.

- (a) *Describe graphs using pictures and set notation.* Ex: Describe C_5 (the cycle on 5 vertices) using both methods.

Solution:

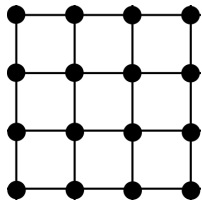
Picture:



Set notation:

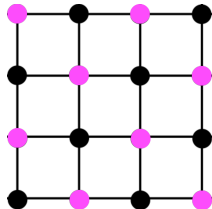
$$G = (\{1, 2, 3, 4, 5\}, \{\{1, 2\}, \{2, 3\}, \{3, 4\}, \{4, 5\}, \{5, 1\}\})$$

- (b) *Find/compute the maximum size clique and/or independent set.* Consider the 4×4 lattice of points below. What is the maximum size clique and maximum size independent set for this graph?



Solution:

Max clique: No triangles, so 2 is max size. Any pair of adjacent vertices forms a maximum clique.
 Max independent set: Color like checkerboard, all vertices of same color are independent (e.g., pink colored set below, or black colored set below). Size 8



- (c) *Compute subgraphs by deleting edges and/or vertices.* Ex: What is $K_{10} - v$, for any vertex v of K_{10} ?
 How many induced/spanning subgraphs does P_{10} have?

Solution:

$K_{10} - v$, for any vertex v of K_{10} is K_9 .

P_{10} has $2^{10} - 1$ induced subgraphs. Recall: for an induced subgraph you delete a subset of the vertices, except you can't delete all vertices or you will get an empty graph. P_{10} has 10 vertices so there are $2^{10} - 1$ different subsets we can delete.

P_{10} has 2^9 spanning subgraphs. Recall: for a spanning subgraph you delete any subset of the edges. P_{10} has 9 edges so there are 2^9 different subsets we can delete.

- (d) *Various counting questions.* Ex: How many cut edges are there in K_9 ? C_9 ? P_9 ? A tree with 9 vertices? A forest with 9 vertices and 4 components? A 5-regular graph on 9 vertices? A binary tree of height 9?

Solution:

K_9 : 0

C_9 (cycle with 9 vertices): 0

P_9 (path with 9 vertices): 8

Tree with 9 vertices: 8

Forest with 9 vertices and 4 components: $9 - 4 = 5$

5-regular graph on 9 vertices: doesn't exist

Binary tree of height 9: anywhere from 9 to $2^{10} - 2$

JUSTIFICATION

There will be two justifications on the exam from the following list. In the case of an if-and-only-if statement, only one direction might be requested. Justifications are usually 2–4 sentences long, sometimes with pictures; they should be logical and clear and give the main idea(s) but aren't necessarily expected to have the formality of a proof.

Modular arithmetic & RSA

- If a is relatively prime to n^{**} , then $ab = ac \pmod n$ implies $b = c \pmod n$ (“Cancellation”).
- If a is relatively prime to n^{**} , then $ab = 0 \pmod n$ implies $b = 0 \pmod n$ (special case of above).

**You can assume this means a has an inverse mod n ; this was proved in class before we got to these statements.

Graphs

- (Thm 47.5) The sum of the degrees of the vertices in G is twice the number of edges, i.e.,
$$\sum_{v \in V} d(v) = 2|E|.$$
- (Lemma 49.7) If G has a walk from x to y then it has a path from x to y .
- (Thm 50.4) A graph is a tree if and only if it has a unique path between any pair of vertices.
- (“cut edge lemma”) If a graph G has a cut edge, the cut edge cannot be on a cycle.
- (Thm 50.5) A **connected** graph is a tree if and only if every edge is a cut edge.
- (Thm 50.7) Every tree with at least 2 vertices has a leaf.
- (Prop 50.8) A tree minus a leaf is still a tree.

PROOF BY INDUCTION

There may be one proof by induction on the exam, one of the following:

- (Thm 50.9) A tree with n vertices has $n - 1$ edges (using the “remove a leaf” induction proof)
- (Thm 50.9) A tree with n vertices has $n - 1$ edges (using the “cut edge” strong induction proof)
- A perfect binary tree of height n has _____ nodes.

FILL-IN-THE-BLANK PROOF

One of the following proofs (or possibly one direction of an if-and-only-if proof) will be given on the exam, mostly already written but with some blanks for you to fill in to show you understand the idea of the proof.

Modular arithmetic & RSA

- a has an inverse mod n iff there exist $x, y \in \mathbb{Z}$ such that $ax + ny = 1$.
- There exist $x, y \in \mathbb{Z}$ such that $ax + ny = 1$ iff a is relatively prime to n .
- Thus (putting those together), a^{-1} exists mod n iff a is relatively prime to n .
- (Lemma for FLT) If a^{-1} exists mod n then multiplication by a permutes the elements $\{1, 2, \dots, n - 1\}$.
- (FLT) If $a \not\equiv 0 \pmod p$ (prime) then $a^{p-1} \equiv 1 \pmod p$.
- (Lemma for Euler's) If a^{-1} exists mod n then multiplication by a permutes the elements of $\mathbb{Z}_n^*$.
- (Euler's) If a is relatively prime to n then $a^{\varphi(n)} \equiv 1 \pmod n$.

GRAPHS - VOCABULARY LIST (May not be a complete list)

Graph, vertex/node, edge, adjacent, degree, max/min degree, isomorphic, regular graph, complete graph, subgraph, induced/spanning subgraph, clique, independent set, complement, walk, path, path graph, connected vertices, connected graph, component, cut edge, cut vertex, cycle, cycle graph, acyclic, forest, tree, leaf, binary tree

BONUS QUESTIONS - Exam will include bonus questions based on sections 51, 52 and 53.